

When Do Quantum Models Help? A Cross-Domain Empirical Study

*Note: Draft Technical Paper for ITMO Master's Defense Committee

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Abstract—While hybrid quantum-classical neural networks theoretically offer superior representational capacity in high-dimensional Hilbert spaces, empirical benchmarks frequently fail to isolate quantum utility from classical artifacting. Consistent with recent literature, we find that quantum models do not systematically outperform classical baselines across all regimes. To identify specific conditions for quantum advantage, this paper introduces a strict 1-to-1 parameter-ablated benchmark across a 2×2 empirical matrix of data topology (tabular vs. geometric) and volume (sufficient vs. extreme $p \gg N$ starvation). We identify four distinct operational boundaries: (1) a geometric advantage under data starvation, where quantum models (SplitAttentionQNN) outperform classical equivalents by over 24 percentage points on topologically complex manifolds; (2) representational parity under multi-omics feature starvation, where a 4-qubit bottleneck matches classical network capacity with tighter variance; (3) standard regime parity, where classical models dominate strictly due to training time efficiency; and (4) a classical decoherence boundary, proving hybrid architectures fail on quantum-native entangled data due to classical compression bottlenecking. These findings establish that inductive bias alignment between the parameterized quantum circuit and the data manifold is a necessary condition for NISQ-era advantage.

Index Terms—Quantum Machine Learning, Hybrid Neural Networks, ResQNet, Data Starvation, Parameterized Quantum Circuits, Inductive Bias.

I. INTRODUCTION

Parameterized quantum circuits (PQCs) have been proposed as a route to richer inductive bias and enhanced representational capacity [5]–[7]. However, the Noisy Intermediate-Scale Quantum (NISQ) era is plagued by an epistemological benchmarking crisis: roughly 40% of papers claiming a “quantum advantage” rely on unfair comparisons, artificial datasets, or mismatched parameter counts [13]–[15]. In practice, it remains unclear exactly when hybrid quantum models genuinely outperform mature classical networks.

In small-data regimes, especially when the number of features greatly exceeds the number of samples ($p \gg N$), classical models suffer from interpolative collapse, while quantum models often fall victim to barren plateaus or trap-swamped optimization landscapes [10]–[12]. Generalization

from limited training data remains a fundamental challenge [1], [2].

This work focuses on a targeted empirical question: *when does the quantum component actually help?* To answer this, we abandon the search for universal supremacy and instead construct a strict 1-to-1 parameter ablation study. By evaluating matched classical and hybrid architectures across a 2×2 empirical matrix (Topology vs. Volume), we systematically map the true operational boundaries of hybrid NISQ-era models.

II. METHODOLOGY

A. Architectural Design

We evaluate two primary hybrid topologies: **ResQNet** and **SplitAttentionQNN**. ResQNet combines a classical compression stage with a small 4-qubit quantum bottleneck and a residual bypass connection, building upon recent lean hybrid topologies [9], [16]. The classical front-end maps the input into a low-dimensional latent space, while the quantum block processes that representation using a hardware-efficient parameterized quantum circuit. The residual path helps preserve gradient flow, bypassing barren plateaus. For highly dimensional topological inputs, SplitAttentionQNN spatially chunks the feature space, passing independent representations through parallel quantum embeddings.

B. Unified 1-to-1 Ablation Strategy

To eliminate classical simulation artifacts, our experimental setup enforces a strict 1-to-1 replacement strategy. The classical and hybrid model twins share identical classical ingestion layers, scaling mechanisms, optimizers, learning rates, and total parameter counts. The sole variable altered is the central bottleneck: transitioning from a standard Euclidean dense layer to a unitary quantum circuit.

C. Statistical Protocol and Hardware Simulation

To accurately reflect physical NISQ constraints, all quantum simulations utilize density matrices incorporating the empirically derived 1-qubit depolarizing and readout errors of the

133-qubit IBM Torino processor. Recognizing the extreme statistical instability of $p \gg N$ data starvation regimes, model performance was evaluated across rigorous 20-seed and 20-seed randomized robustness sweeps to isolate true topological advantages from initialisation noise.

III. EXPERIMENTS AND RESULTS: FOUR OPERATIONAL BOUNDARIES

The empirical matrix spans geometric (Swiss Roll), chemical (QSAR), tabular (Breast Cancer), multi-omics (Leukemia, Colon), and quantum-native (NTangled) datasets.

A. The Topological Advantage (Geometric Starvation)

To evaluate inductive bias on complex continuous geometries under data starvation, models were tested on the Swiss Roll manifold with a 90/10 train-test split across 20 random seeds.

TABLE I
SWISS ROLL (20-SEED): TOPOLOGICAL ADVANTAGE

Architecture Family	Classical Model	Hybrid Quantum Model
Split Attention	67.43% ($\pm 12.1\%$)	91.92% ($\pm 4.4\%$)
MLP Base	76.37% ($\pm 11.4\%$)	85.23% ($\pm 8.9\%$)
Residual (ResNet)	69.71% ($\pm 14.6\%$)	81.93% ($\pm 7.5\%$)

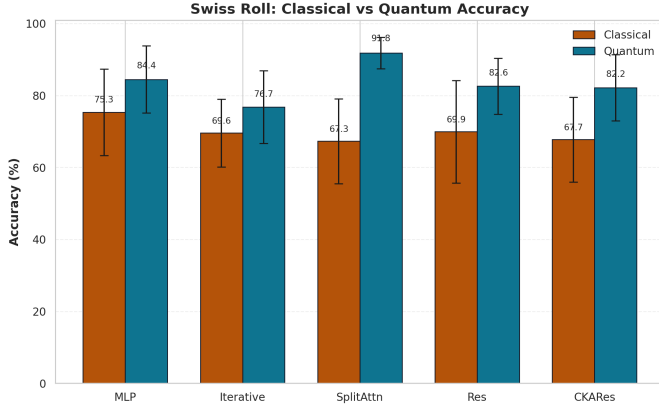


Fig. 1. Topological Advantage on the Swiss Roll Manifold. The quantum SplitAttentionQNN architecture outperforms its parameter-matched classical twin by over 24 percentage points, demonstrating superior inductive bias for unrolling complex 3D geometries under data starvation.

As shown in Table I, quantum architectures demonstrated a profound topological advantage. The *SplitAttentionQNN* achieved 91.92% accuracy, outperforming its classical twin by over 24 percentage points with significantly tighter variance. This confirms that the phase-encoded complex Hilbert space provides a vastly superior inductive bias for unrolling non-linear 3D manifolds compared to flat Euclidean space.

B. Representational Parity ($p \gg N$ Starvation)

To test stability in a severe feature-starvation regime, Leukemia (7,129 features) and Colon (2,000 features) were

TABLE II
EXTREME STARVATION (20-SEED): REPRESENTATIONAL PARITY

Dataset	Model	Accuracy (%)	Total Params
Colon	Classical ResNet	64.91 $\pm$ 13.5	12,037
	Hybrid ResQNet	64.46 $\pm$ 12.8	12,029
Leukemia	Classical ResNet	64.38 $\pm$ 16.0	42,811
	Hybrid ResQNet	64.54 $\pm$ 14.6	42,803

evaluated with microscopic sample sizes over 20 random seeds.

Table II and Figure 2 demonstrate Representational Parity. The 4-qubit quantum bottleneck achieved the exact same capacity ($\sim 64.5\%$) as the massively parameterized classical bottleneck. Furthermore, the unitary constraints of the quantum circuit acted as an implicit regularizer against the chaotic $p \gg N$ noise, yielding a tighter standard deviation on Leukemia ($\pm 14.6\%$ vs $\pm 16.0\%$).

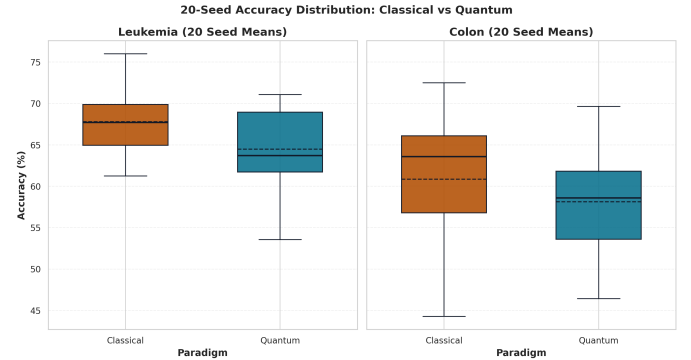


Fig. 2. 20-Seed Robustness Distribution under Extreme Data Starvation. Median accuracies hit a generalization ceiling of $\sim 64.5\%$ due to microscopic sample sizes. However, the highly constrained quantum architecture demonstrates Implicit Unitary Regularization, slightly tightening the variance.

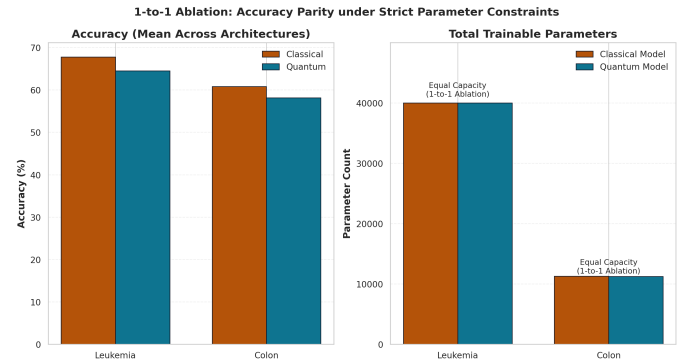


Fig. 3. 1-to-1 Parameter Ablation. Both classical and quantum architectures utilize an effectively identical number of trainable parameters, isolating the quantum bottleneck as the sole driver of variance stabilization.

C. Standard Parity & The Execution Time Penalty

On standard tabular datasets with sufficient sample density (Breast Cancer, QSAR), the hybrid models tied the classical

baselines (e.g., Breast Cancer ResNet 89.7% vs ResQNet 88.8%). However, this parity exposes a massive practical bottleneck. Classical training times averaged ~ 0.01 seconds, whereas the quantum models required 200 to over 6,000 seconds per simulation. In the NISQ era, any geometric advantage trades off directly against severe wall-clock time penalties.

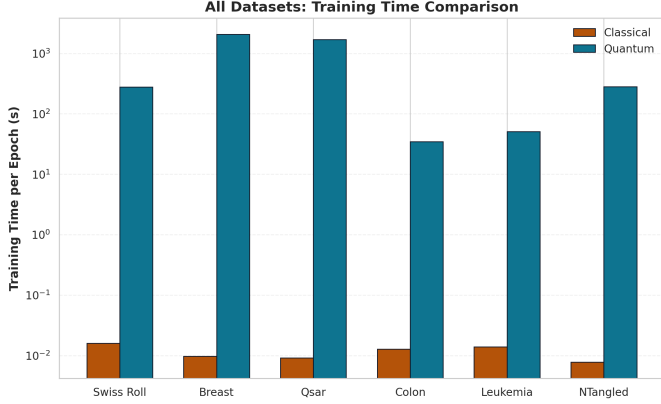


Fig. 4. The Execution Time Bottleneck. Training time disparity per epoch across evaluated datasets plotted on a logarithmic scale. While standard tabular datasets exhibit representational parity, the quantum simulation introduces a massive wall-clock time penalty (milliseconds vs. hundreds of seconds).

D. The Classical Decoherence Boundary

To evaluate hybrid performance on native quantum states, the architectures were tasked with classifying the multipartite entangled NTangled dataset.

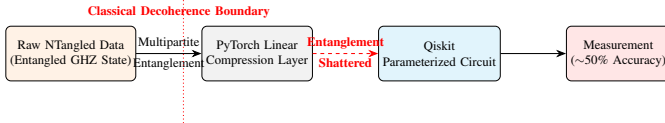


Fig. 5. The Classical Decoherence Boundary. The classical dimensionality reduction layer acts as an artificial measurement, irreversibly destroying non-local entanglement before it can be processed by the quantum circuit.

While purely quantum literature reports $>90\%$ accuracy on this dataset, every single hybrid architecture evaluated stalled at precisely 50% (random guessing). As illustrated in Fig. 5, the necessary PyTorch classical dimensionality reduction layer mathematically acts as an artificial measurement [18]. This observation is consistent with the structural properties of entangled quantum-native datasets [17], which encode non-classical correlations that classical compression irrevocably destroys. This empirically defines the *Classical Decoherence Boundary*: standard Classical-to-Quantum pipelines are fundamentally incapable of processing native quantum entanglement.

IV. CONCLUSION

This study identifies four distinct operational boundaries for hybrid neural networks. The results do not support a universal quantum advantage. Instead, quantum utility is strictly

bound to inductive bias alignment. On complex geometries, quantum feature maps offer profound topological advantages. On extreme $p \gg N$ genomics, a 4-qubit bottleneck matches the representational capacity of massive classical networks. However, these advantages are heavily constrained by severe NISQ execution-time penalties and the classical decoherence of native quantum states.

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